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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Comparative Study of NLP Models and Algorithms

DSA0305-Natural Language Processing

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Introduction to NLP

Why comparing models and algorithms matters

1

What is NLP?

A field of AI that enables machines to read, interpret, and generate human language.

2

Why compare models?

Different tasks (translation, sentiment, Q&A) need different trade-offs in accuracy, speed and data needs.

3

Evolution

NLP has moved from hand-written rules to statistics, then deep learning, then transformers.

4

This study

Benchmarks 3 model families on accuracy, complexity, interpretability and compute cost.



Traditional & Statistical Models

Rule-based and probabilistic approaches

- | | | | |
|---|--|---|---|
| 1 | Rule-Based Systems | Hand-crafted grammar & pattern rules (e.g. regex, ELIZA) | <i>High interpretability, no training data needed</i> |
| 2 | Naive Bayes | Probabilistic classifier using word-independence assumption | <i>Fast, strong baseline for text classification</i> |
| 3 | Hidden Markov Models (HMM) | Sequence labelling via state transition probabilities | <i>Used in POS tagging, speech recognition</i> |
| 4 | Conditional Random Fields (CRF) | Models dependencies between neighboring labels | <i>Stronger than HMM for NER, chunking</i> |



Machine Learning & Deep Learning Models



Feature-based learning to neural sequence models

Classical Machine Learning

- SVM — strong margin-based text classifier, works well on small/medium datasets
- Random Forest — ensemble of decision trees, robust to noisy features
- Logistic Regression — simple, fast, interpretable linear baseline
- Requires manual feature engineering (TF-IDF, n-grams)

Deep Learning

- RNN — processes sequences step-by-step, captures short-term context
- LSTM / GRU — gated memory cells solve vanishing-gradient, capture long-range dependencies
- CNN for text — extracts local n-gram-like features via convolution filters
- Learns features automatically, needs larger labelled datasets

Transformer-Based Models

B

BERT

Bidirectional encoder, pre-trained via masked-language modeling

Best for: Classification, NER, Q&A

G

GPT (3/4 family)

Autoregressive decoder, generates text left-to-right

Best for: Text generation, chat, summarization

T

T5

Text-to-text unified framework, encoder-decoder

Best for: Translation, summarization, multi-task



Comparative Analysis

Accuracy, complexity, interpretability and cost

Model Family	Typical Accuracy	Training Cost	Interpretability	Data Needed
Rule-Based	Low–Medium	None	Very High	None
Statistical (HMM/CRF)	Medium	Low	High	Small
Classical ML (SVM/RF)	Medium–High	Low–Medium	Medium	Medium
Deep Learning (RNN/LSTM)	High	High	Low	Large
Transformers (BERT/GPT)	Very High	Very High	Very Low	Very Large



Conclusion & Future Directions

1

Choice depends on task, data availability, interpretability needs and compute budget.

2

Progress moves from rule-based → statistical → deep learning → transformers, trading interpretability for accuracy.

3

Efficient transformers, few-shot learning, and hybrid symbolic-neural systems to cut cost and improve reasoning.



THANK YOU